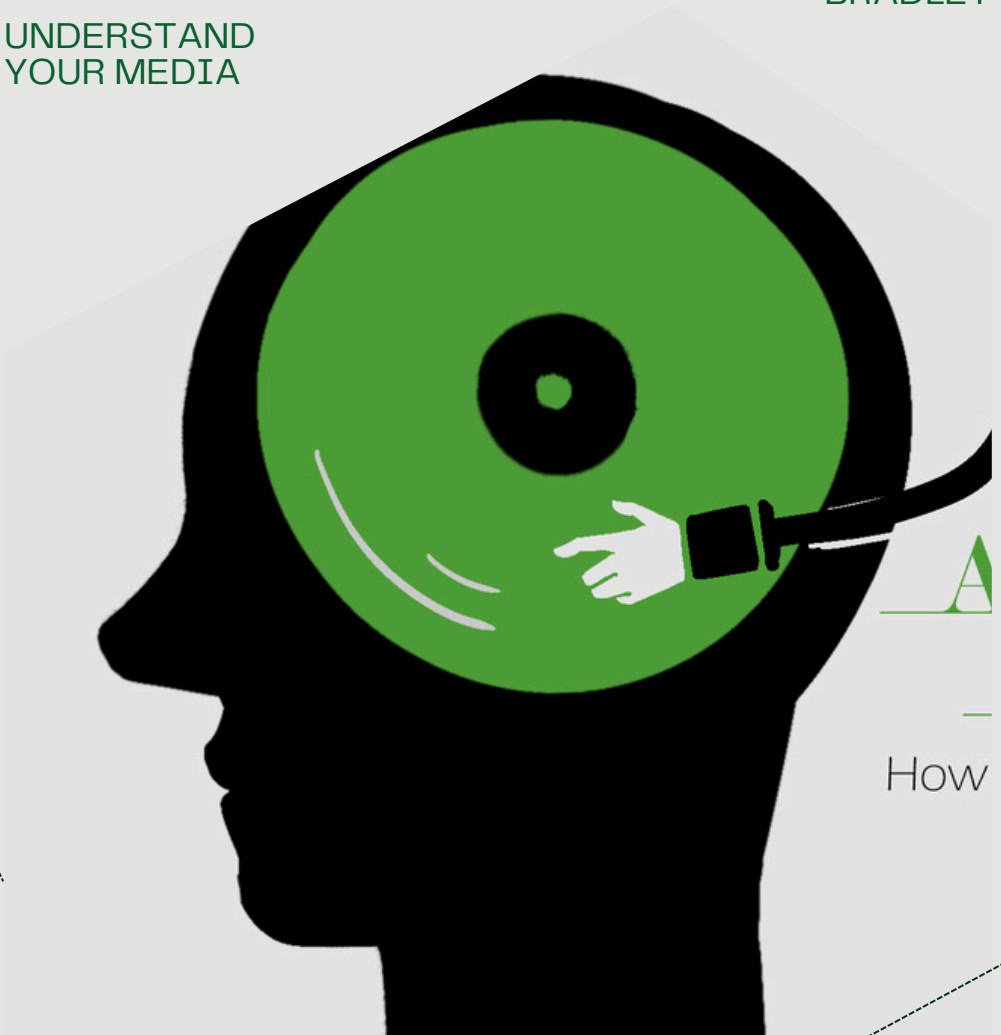
SPOTIFY ALGORITHMIC RESEARCH

BJORN
BRADLEY

UNDERSTAND
YOUR MEDIA



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How

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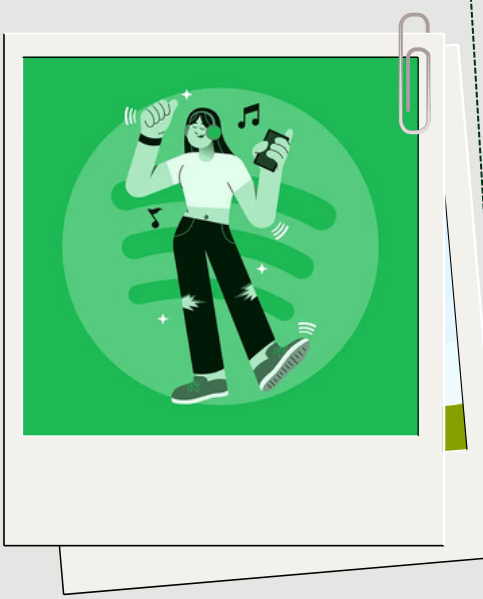


Algorithmic System Description

Spotify's recommendation system is a large-scale personalization engine that predicts what audio content—songs, albums, playlists, podcasts, and mixes—a listener is most likely to engage with, and then organizes that content across recommendation surfaces like Home, Radio, Autoplay, and personalized mixes. It is used primarily by listeners through everyday platform behavior: creating an account, searching, listening, skipping, saving to a library, and following artists. Spotify describes these ongoing actions as signals that shape a user's "taste profile," which the system uses to interpret what a person is interested in and how they prefer to listen.

Beyond individual behavior, the system also depends on aggregate patterns across users: when many people interact with similar content in similar ways, Spotify treats those trends as evidence to improve recommendations for everyone, especially in search and discovery contexts. Alongside these behavioral and trend signals, Spotify integrates human oversight through editorial curation and "listener safety" practices that can influence what is promoted, what is contextually placed, and what is restricted from recommendation. Finally, Spotify includes an artist-facing lever—Discovery Mode—where artists or labels can add a promotional signal to increase the likelihood a track appears in certain personalized listening contexts, with Spotify taking a commission on royalties in those contexts; this inserts commercial influence directly into the recommendation pipeline.

System Diagram

Data Input

User Input

- Plays
- Skips
- Likes
- Follows

Artist Input

- Track Uploads
- Discovery Mode Opt-in
- Meta Data

Trend Signals

- Aggregate Listening Patterns
- Viral Spikes
- Regional Trends

Audio Analysis

- Tempo
- Energy
- Danceability
- Valence

Data Processing

- Log aggregation
- Interaction Datasets
- Data Cleaning
- Bot Filtering
- Feature Encoding
- Embedding generation

Recommendation Model

Generation

- Collabrative Filtering
- User Similarity
- Track Similarity
- Song Pool

Ranking

- Relevance Prediction
- Engagement Likelihood
- Song Prioritization
- Personalized Ordering

Constraint

Editorial Curation

- Human Playlist Creation
- Cultural Trend Knowledge
- Playlist Placement

Platform Governance

- Content Moderation
- Safety Policies
- Recommendation Restrictions

Promotional Prioritization

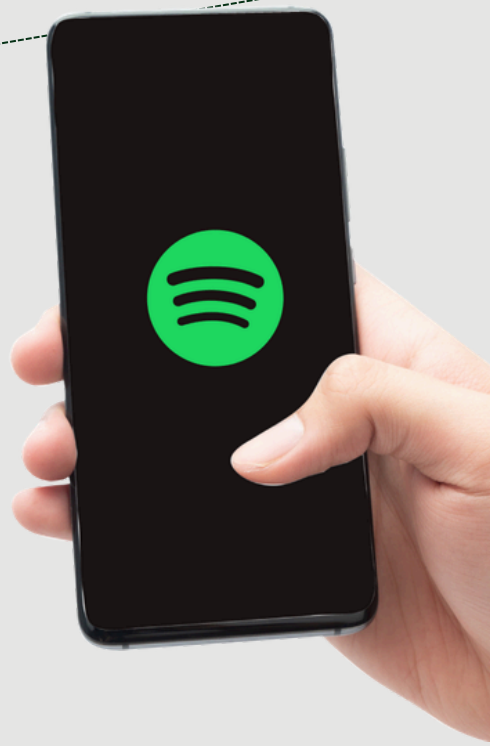
- Discovery Mode weighting
- Artist marketing signals

Recommendation Output

- Discover Weekly
- Daily Mix
- Autoplay
- Radio
- Home Screen
- Niche Mixes
- Enhanced Playlist Mode

Data Collection & Usage

Spotify's recommendation system relies on three primary categories of data: behavioral interaction data, audio-derived content features, and structured metadata. Together, these inputs allow the platform to model user preference and predict future listening behavior.



First, the system collects behavioral data generated through everyday use. This includes plays, skips, replays, search queries, playlist additions, follows, and listening duration. Spotify's Privacy Policy confirms that it gathers information about how users interact with the service, including usage and device data (Spotify Privacy Policy). These signals are aggregated across millions of users and analyzed using collaborative filtering techniques, which identify patterns by comparing similar listening behaviors at scale (Koren, Bell, & Volinsky, 2009). In this framework, a listener's "taste profile" is shaped statistically through collective behavior.

Second, Spotify incorporates content-based audio features extracted from tracks. Through automated analysis, songs are translated into measurable attributes such as tempo, valence, energy, and danceability (Spotify Web API Documentation). These numerical representations allow the system to compute similarity between tracks, supplementing behavioral modeling with intrinsic properties of the audio itself (Ricci et al., 2011).

Data Construction & Processing

Category	Example	Purpose
User		
Technical Usage		
General Usage		
Device Sensor		
Voice & Message		
Ad. Personal		
Third Party		
Street Address		
Payment		

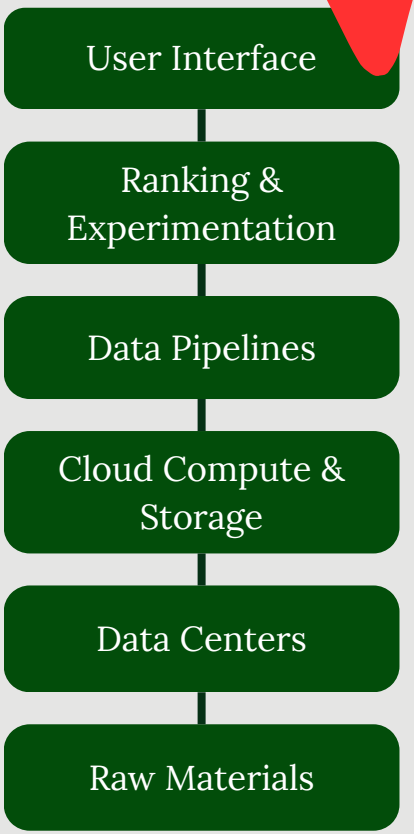
The data that powers Spotify's recommendation models does not exist in a finished form; it must be continuously constructed, filtered, and structured. Every listening event generates a log entry that includes user ID (anonymized), track ID, timestamp, device type, session duration, and interaction outcome (e.g., skip, replay, save). These raw logs are aggregated at scale and transformed into structured training datasets.

Preprocessing involves normalizing listening duration, filtering bot or fraudulent streams, encoding categorical variables such as genre and artist, and generating embeddings that compress high-dimensional interaction data into usable vector representations. Recommender system research emphasizes that this transformation stage is critical, as design choices in filtering and weighting directly influence model outputs (Ricci et al., 2011).

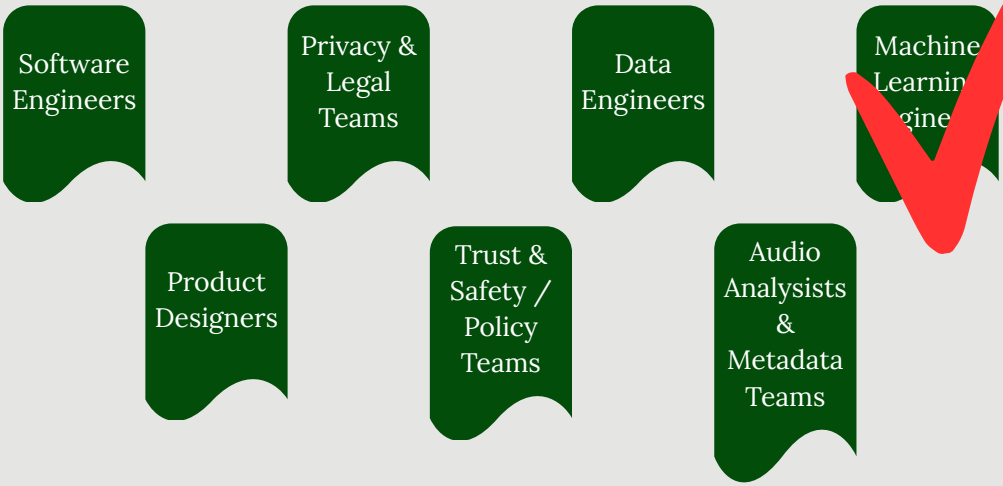
Dataset distribution also reflects structural imbalance. Popular tracks generate disproportionately more interaction data, reinforcing their visibility through feedback loops commonly described as popularity bias (Koren et al., 2009). As a result, the training corpus is not neutral; it reflects existing listening hierarchies and platform dynamics. The dataset is continuously updated, meaning the system retrains on patterns that are themselves shaped by prior recommendations.

Infrastructure

Spotify's recommendation system operates on a layered material infrastructure that extends far beyond the interface seen by listeners. At the surface, users encounter playlists, autoplay suggestions, and personalized mixes. Beneath this interface sits a ranking system that continuously evaluates engagement metrics, tests model variants through experimentation, and updates recommendation weights. These processes rely on extensive data pipelines that log user interactions, clean and normalize behavioral signals, and encode them into structured inputs for machine learning models.



This computational layer depends on distributed cloud infrastructure, including large-scale storage systems and high-performance computing resources capable of processing billions of listening events. These systems are housed in data centers that require constant electricity and water for cooling, embedding music recommendation within global energy and resource networks. The algorithm is therefore not an abstract entity but a material system sustained by servers, networking hardware, and physical facilities. Even at its lowest level, this infrastructure relies on mineral extraction, hardware manufacturing, and global supply chains. Personalization is built upon layered technological and environmental foundations that make real-time prediction possible.

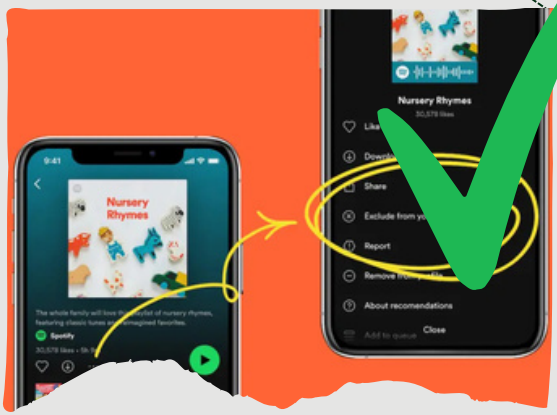


The recommendation system is sustained not only by hardware and code, but by layered forms of human labor. Software engineers design the logging mechanisms that capture user interaction data, while data engineers structure and maintain the large-scale datasets required for model training. Machine learning engineers develop ranking models and continuously refine them through experimentation and performance evaluation. Audio analysis teams and metadata specialists contribute structured descriptors that translate sound into measurable attributes.

Beyond technical development, legal and privacy teams establish compliance frameworks that govern how user data may be collected and processed. Trust and safety teams interpret community guidelines and determine how content is surfaced, restricted, or contextualized. Product designers shape the recommendation surfaces through interface decisions that influence visibility and engagement. These forms of labor collectively determine how personalization operates in practice.

At the same time, certain forms of work remain less visible, including contractual moderation, support operations, and the labor of artists whose music becomes training data for recommendation models. The algorithm is therefore not an autonomous decision-maker, but a socio-technical system shaped by institutional priorities, legal constraints, and organizational hierarchies. Personalization emerges from coordinated technical, managerial, and regulatory labor embedded within a global streaming economy.

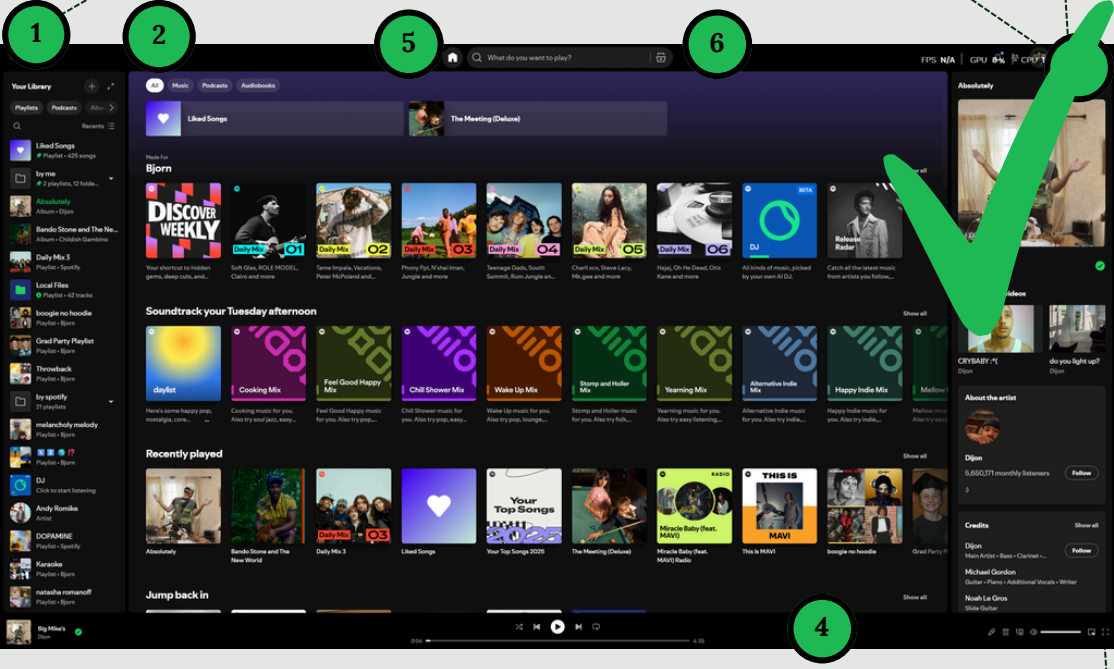
Public Profile



Spotify publicly frames its recommendation system as a tool for personalized discovery and creative connection. Across its interface and promotional materials, personalization is presented as intuitive and user-centered. Features such as Discover Weekly, Daily Mixes, and “Made For You” playlists are described as uniquely tailored to individual taste profiles, shaped by listening behavior and engagement patterns. In its public explanation of recommendations, Spotify states that it analyzes user interactions, listening history, and broader listening trends to surface relevant content (Spotify, “Understanding Recommendations”). The dominant narrative emphasizes convenience: the system learns what a user likes and delivers music seamlessly, reducing friction in discovery.

At the same time, Spotify positions its algorithm as a mechanism for artist opportunity. Corporate messaging highlights the platform’s role in connecting creators with audiences, often describing its recommendation tools as amplifying emerging artists and democratizing access to listeners. Products such as Discovery Mode are framed as ways for artists to increase exposure within personalized listening contexts. Through this language, Spotify presents itself as a neutral intermediary that facilitates cultural exchange rather than directs it.

This public framing foregrounds personalization, empowerment, and access while backgrounding the technical and economic mechanisms that govern ranking and visibility. The recommendation system is described as responsive and adaptive, yet its internal weighting processes, promotional incentives, and optimization metrics remain largely abstracted from public view. As a result, Spotify’s corporate narrative constructs personalization as both intimate and democratic, positioning the algorithm as a benevolent guide within the digital music ecosystem.



1

This left pane holds the users library. This can be sorted by category or filter, but consists of all liked or created content such as liked songs, playlists, albums, artists, etc. Each of these entities can be selected to open and play or right clicked to edit

2

This is the main section of the page consisting of all the recommended content. In field notes I go further in depth (not enough room here) but the idea is Spotify gives you different sections all housing created or generated playlists. The only content created by me would be in recently played

3

This section houses data regarding the track currently playing. For example here I am playing a song by childish gambino so it shows the song, related videos, content about the artist, and credits. There are also buttons to follow the artist or like the track

4

This section similar to 3 also holds track data but instead regards the playing of that data. To the left is the image, title, artist name, and a button to like and then you have your typical shuffle, back, play, forward, and repeat with a timeline. Then you have additional buttons to go into further track data such as lyrics

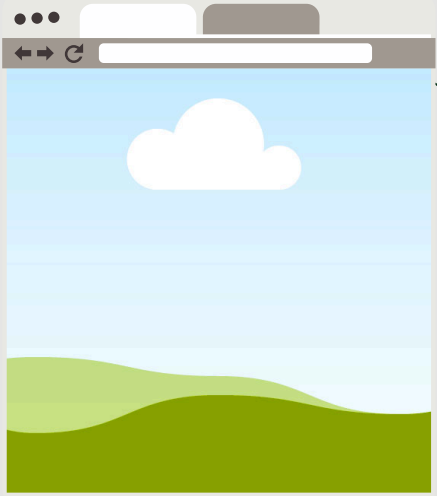
5

Here we have the search bar. The search bar obviously allows for search functionality but also when selected changes the whole page instead of a drop down to give maximum recommendation. Additionally you are not just searching for a track but anything in spotify and all categories possible will be shown in recommendation

6

Finally we have a button to take the user to the explorer page where they are met with different categories to find more created or generated playlists, podcasts, etc.

User Experience



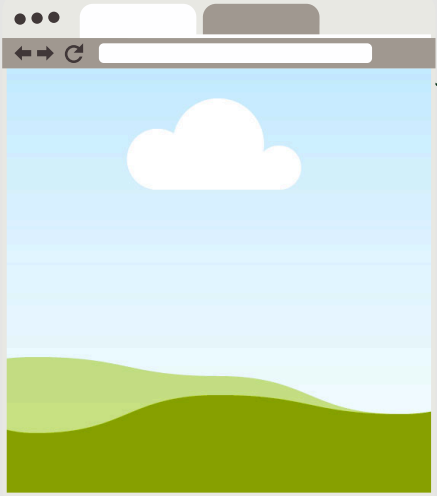
Spotify's user experience begins with a visually consistent and highly recognizable interface. The platform uses a dark background paired with green accent colors, rounded containers, and large album artwork to organize content in a way that feels both simple and visually engaging. These aesthetic choices create a sense of familiarity and comfort while allowing a large amount of music data to be displayed without overwhelming the user.

The home page functions as the central hub of interaction, displaying recommended playlists and listening categories such as "Recently Played," "Your Top Mixes," and "Inspired by Your Recent Activity." From this starting point, users can navigate to different sections of the platform through the library panel, search tools, or recommendation shelves.

Searching for music produces a particularly distinctive interaction. Instead of presenting a simple dropdown, Spotify reorganizes the entire page to display search results grouped into categories including top results, songs, artists, albums, playlists, and genres. This layout allows users to explore music across multiple formats while maintaining a consistent visual structure.

Overall, Spotify emphasizes clarity and familiarity in its design. Navigation elements remain consistent across pages, playback controls remain visible at all times, and large visual containers guide the user's attention toward recommended music.

User Experience



Spotify's recommendation system becomes most visible through everyday listening interactions. Users continuously provide feedback to the platform through actions such as playing songs, skipping tracks, liking music, searching for artists, or adding songs to playlists. These behaviors act as signals that inform Spotify's recommendation system and shape future listening suggestions.

While the system rarely explains exactly how these signals influence recommendations, many relationships between user behavior and algorithmic response can be inferred. For example, repeatedly playing certain artists or songs often leads to similar music appearing in recommended playlists, radio sessions, and autoplay sequences. Users may therefore learn to influence the system indirectly through their listening habits.

Despite this feedback loop, the algorithm itself remains largely invisible. Users interact with playlists and recommendations without seeing how their data is processed or how songs are prioritized within recommendation models. As a result, Spotify's user experience blends transparency and opacity: listeners clearly see the outputs of the algorithm but have limited insight into how those recommendations are produced.

This design allows the platform to maintain a highly personalized experience while quietly collecting behavioral data that continuously trains the recommendation system.

Discovery Mode & Pay-for-Visibility

Introduction

Spotify's Discovery Mode is an artist-facing feature that allows artists or labels to increase the likelihood that selected tracks appear within certain algorithmic recommendation contexts, including Radio and Autoplay. In exchange, Spotify applies a commission to recording royalties generated from those streams. According to Spotify's documentation, "a 30% commission is applied to recording royalties generated from all streams of selected songs… in Discovery Mode contexts"

Because recommendation surfaces such as personalized mixes, radio, and autoplay are major channels through which listeners encounter music, placement within these systems is economically and culturally significant. I use the term pay-for-visibility to describe systems in which promotional exposure is exchanged for financial value within algorithmic environments. Discovery Mode raises ethical concerns because it embeds financial incentives directly into recommendation ranking.

Thesis: Spotify's Discovery Mode creates an ethically problematic pay-for-visibility system by integrating financial compensation into algorithmic recommendation. This structure introduces economic pressure on artists, obscures promotional influence from listeners, and reinforces platform control over visibility.

Pay for Visibility

Discovery Mode operates by increasing the probability that selected tracks are surfaced in algorithmic recommendation contexts. In return, Spotify takes a percentage of royalties generated from those streams. Although artists are not paying upfront fees, the commission structure functions as a financial exchange tied to algorithmic exposure.

Scholarly work on streaming payola highlights how online platforms avoid traditional broadcast regulation. As noted in legal scholarship, streaming platforms do not fall under FCC jurisdiction, meaning algorithmic pay-for-play operates with fewer federal restrictions. Together, Spotify's documentation and regulatory scholarship demonstrate that Discovery Mode converts recommendation into a monetized promotional channel.

Hidden Influence

Discovery Mode functions within recommendation surfaces that appear organic to listeners. There is no visible indicator on the user interface that a track's placement may be financially influenced. This lack of disclosure blurs the distinction between neutral personalization and paid promotion.

Research on platform visibility systems argues that algorithmic ranking often reflects commercial priorities even when framed as neutral recommendation. When promotional signals are embedded within systems labeled as "personalized," users are unable to distinguish between behavioral relevance and economic influence. Discovery Mode therefore introduces commercial weighting without clear labeling.

Control

Spotify controls eligibility requirements, ranking systems, and performance measurement for Discovery Mode . Scholars have war that pay-for-play systems may concentrate power among dominant platforms, potentially narrowing diversity and disadvantaging smaller artists . By defining who qualifies, how exposure is measured, and how royalties are adjusted, Spotify positions itself as the central arbiter of musical visibility.

Conclusion & Significance

Together, these points demonstrate that Discovery Mode transforms Spotify's recommendation from a neutral personalization tool into a monetized promotional infrastructure. By embedding financial trade-offs into ranking systems, Spotify reinforces platform power while reshaping how cultural exposure is distributed in digital music environments.

Spotify Personalization, Data, & User Autonomy

Introduction

Spotify presents personalization as a feature that improves user experience by recommending music tailored to individual taste. Features such as Discover Weekly, Daily Mixes, and Autoplay are powered by machine learning systems that analyze listening behavior, including skips, saves, searches, playlist additions, and listening. Spotify publicly states that its system adapts in real time to user activity, effectively creating a distinct listening environment for each individual.

However, personalization depends on continuous behavioral tracking and algorithmic profiling that most users cannot fully observe or understand. Scholars studying digital platforms and recommender systems have examined how such systems extract behavioral data, shape user autonomy through feedback loops, and concentrate informational power within opaque infrastructures.

Thesis: Spotify's personalization system raises ethical concerns because it relies on continuous behavioral data extraction, constructs predictive profiles that can steer user autonomy, and concentrates informational power through algorithmic opacity.

Continuous Behavioral Data Extraction

Spotify's personalization depends on detailed behavioral monitoring. According to Spotify's Privacy Policy, personal data is "collected and processed" when users access or use the service. This confirms that personalization is inseparable from ongoing data collection embedded within ordinary listening.

Scholarly analysis of Spotify's evolving privacy documentation shows that its data practices have expanded over time, reflecting increasing complexity in how user data is collected and modeled. Together, these sources demonstrate that personalization requires sustained data extraction. Participation in the platform becomes conditional on allowing continuous monitoring, normalizing behavioral surveillance as a prerequisite for digital cultural access.

Profiling & Steering User Autonomy

Spotify does not simply reflect user preferences; it constructs predictive taste profiles that shape future exposure. Research explains that Discover Weekly combines individual listening data with patterns derived from other users' behaviors. This hybrid approach integrates collaborative filtering, contextual modeling, content-based analysis, and natural language processing to assemble detailed listener profiles.

Because these profiles are fed back to users through repeated recommendation cycles, the system can gradually influence what feels visible, relevant, and desirable. Over time, feedback loops may reinforce certain genres, moods, or artists, shaping listening habits rather than merely reflecting them. Personalization thus becomes a steering mechanism that subtly structures cultural exposure.

Opacity & Informational Power

Discovery Mode operates by increasing the probability that selecu tracks are surfaced in algorithmic recommendation contexts. In reu Spotify takes a percentage of royalties generated from those streams Although artists are not paying upfront fees, the commission structure functions as a financial exchange tied to algorithmic exposure.
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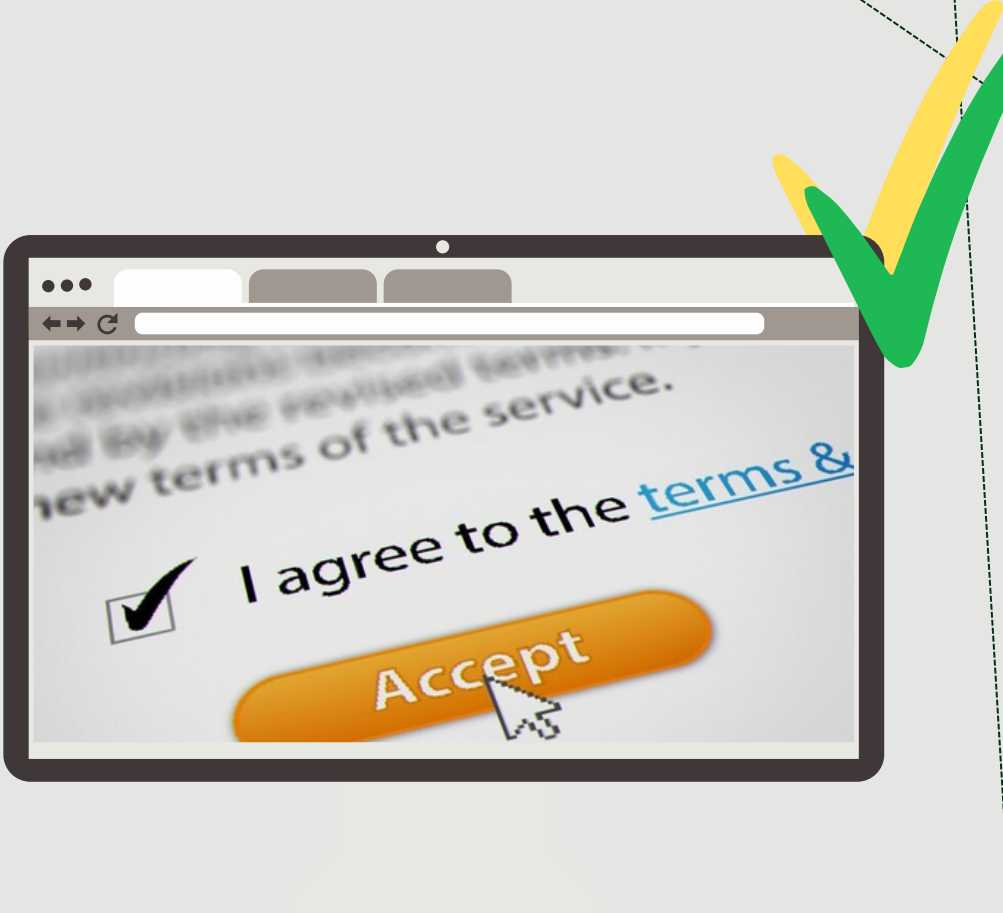
Transparency

Subheading Here

Category	Explanation	Score
Policy	Clarity of publications and disclosures regarding rules and guidelines	4
Algorithmic	Level of transparency regarding recommendations and the algroithm within	3
Data	How clearly Spotify discloses what data is being collected, why, and what it's used for.	4
User Control	Level of control users have over personalization and whether its explained	3

3.5

Spotify provides substantial public documentation explaining its rules, personalization framework, and data practices. At a categorical level, transparency appears strong. Policies outline prohibited behavior, enforcement authority, data collection categories, and general explanations of recommendation systems. However, transparency weakens when moving from descriptive language to operational mechanics. While Spotify explains what influences recommendations and acknowledges commercial considerations in content placement, it does not disclose ranking weights, optimization metrics, or the measurable impact of promotional mechanisms. This section examines the distinction between procedural transparency—what categories exist—and operational transparency—how decisions are computed and enforced in practice.



Spotify's User Guidelines and End User Agreement clearly articulate prohibited conduct and enforcement rights. The platform reserves authority to remove content, suspend accounts, and monitor activity. Notably, the End User Agreement acknowledges that content selection and placement "may be influenced by commercial considerations." This statement signals that recommendation is not exclusively behavioral. However, Spotify does not disclose how commercial influence interacts with algorithmic ranking, nor does it quantify the magnitude of promotional adjustments. Enforcement thresholds, appeal success rates, and audit metrics are also not publicly detailed. Policy transparency is therefore strong at the level of rules and disclosure, but limited in explaining how governance decisions translate into measurable ranking outcomes.

Understanding recommendations on Spotify

How do Spotify recommendations work?

At Spotify, we aim to create great and unique experiences for each user. Our goal is to connect everyone with what they love and h hem discover something new. No two listeners are the same, so everyone's Spotify experience, and many of our recommendations, are personaliz hen asked what they like about Spotify, most listeners cite our personalization as their top feature. You might wonder how we generate these recommendations across the Home feed, playlists, search results or other parts of the service, and we want to help demystify how they

At Spotify, people and technology work together to deliver relevant recommendations. Some recommendations are based on editorial cu a pop playlist created by music editors. Other recommendations are tailored to each listener's unique taste, like a personalized playlist pow our expert-designed algorithms.

We believe recommendations shouldn't just optimize for the next click, but evolve with your taste. We have dedicated teams in place making su your recommendations enable real engagement and cultivate meaningful connections. We're always working on improving our recommendation systems to make sure we're showing you relevant and enjoyable content.

Editorial curation

Editors at Spotify use data insights, sharp ears, and an understanding of cultural trends to place content where it's likely to resonate most with fans around the world. They employ thoughtful curation to recommend content on Spotify, like in editorial playlists. Around the world, editors at Spotify have extensive knowledge about local music and culture, which allows them to make their programming decisions with the best listener experiences in mind.

Personalized recommendations

Spotify offers algorithmic recommendations that are relevant, unique, and specific to each user. Our algorithms select and order content across each listener's Spotify experience, including in Search, Home, and in personalized playlists.

To make these recommendations, our algorithms rely on a number of inputs. The importance of these inputs may vary over time, depending on your individual use of Spotify. We believe that your taste profile is the most important input for creating the best overall user experience. Below, you'll find more details on the most significant inputs and how they work.

Your "taste profile"

As you engage with Spotify, actions such as searching, listening, skipping, or saving to Your Library influence our interpretation of your taste. We call this your "taste profile," and it gives our algorithms an indication of what you're interested in and how you like to listen.

- Example: If you listen to a certain artist, we might recommend more songs by that artist.
- Example: Our Release Radar playlist recommends the latest releases we think you'll enjoy based on similar music you've listened to.
- Example: If you listen to a sports podcast, we might recommend other sports podcasts to you.

Information you share with us

Recommendations are also based on information you share with Spotify, like your general (non-precise) location, your language, your age, and who you follow. This gives our algorithms signals about what topics you're interested in or which artists you want to keep up to date with.

- Example: If you follow a certain podcast, we might recommend an episode from that podcast.
- Example: If you choose German as your language on Spotify, we might recommend German-speaking podcasts.

Spotify's "Understanding Recommendations" page provides a conceptual overview of personalization. It explains that recommendations draw from listening history, skips, saves, follows, and broader trends, combining editorial and algorithmic systems. This disclosure clarifies inputs but not computation. Spotify does not reveal ranking weights, model evaluation criteria, how conflicting signals are resolved, or how long historical data influences future recommendations. While users are informed that behavior shapes personalization, they cannot directly inspect, edit, or reset their "taste profile." Data transparency exists at the categorical level, but operational transparency remains limited. Users receive explanations of influence without visibility into the mechanisms that convert influence into ranked exposure.

Primary Sources

- Spotify. “Understanding Recommendations on Spotify.” Spotify.com, 2024, www.spotify.com/us/safetyandprivacy/understanding-recommendations.
- “Privacy Policy - Spotify.” Spotify.com, 2024, www.spotify.com/us/legal/privacy-policy.
- “Discovery Mode – Spotify for Artists.” Spotify.com, 2024, artists.spotify.com/discovery-mode?
- Spotify. “Spotify.” Spotify, open.spotify.com/.
- https://www.spotify.com/us/safetyandprivacy/transparency
- Spotify. “Listening Is Everything.” Www.spotify.com, www.spotify.com/us/legal/user-guidelines/.
- Spotify. “Music for Everyone.” Spotify.com, 2019, www.spotify.com/us/legal/end-user-agreement/.

Secondary Sources

1. https://journals.sagepub.com/doi/full/10.1177/14614448211022161?_gl=1*19dt56r*_up*MQ..*_ga*MTM3ODA4MjU3OC4xNzcwMjU1NzY3*_ga_60R758KFDG*czE3NzAyNTU3NjYkbzEkZzEkdDE3NzAyNTU3ODUkajQxJGwwJGgxNDUwNTk4MDcy
2. https://journals.sagepub.com/doi/full/10.1177/0163443717745147?_gl=1*1oarqlr*_up*MQ..*_ga*MTM5OTcxOTU5Ny4xNzcwMjU1OTAw*_ga_60R758KFDG*czE3NzAyNTU5MDAkbzEkZzEkdDE3NzAyNTU5NDYkajE0JGwwJGgxODAxMDkyNDk5
3. https://firstmonday.org/ojs/index.php/fm/article/view/11783
4. https://www.tandfonline.com/doi/full/10.1080/1369118X.2020.1736124
5. https://journals.sagepub.com/doi/full/10.1177/13548565241277379?_gl=1*1hm4uk2*_up*MQ..*_ga*MTQ2NDUzNDQ3NC4xNzcwMjU2MTMz*_ga_60R758KFDG*czE3NzAyNTYxMzMkbzEkZzEkdDE3NzAyNTYxNDMkajUwJGwwJGgxODcwMTQ4MTQw
6. https://journals.sagepub.com/doi/full/10.1177/2053951720923377?_gl=1*109ol95*_up*MQ..*_ga*NzI1MTgxMjc4LjE3NzAyNTYwNjI.*_ga_60R758KFDG*czE3NzAyNTYwNjEkbzEkZzEkdDE3NzAyNTYwNjckajU0JGwwJGgyOTQ4OTA2OTY.
7. https://scholarship.law.duke.edu/cgi/viewcontent.cgi?article=6825&context=faculty_scholarship
8. https://firstmonday.org/ojs/index.php/fm/article/view/11783/11150
9. https://scholarship.law.duke.edu/cgi/viewcontent.cgi?article=6825&context=faculty_scholarship
10. https://pmc.ncbi.nlm.nih.gov/articles/PMC11459776
11. https://ebookcentral-proquest-com.ezproxy1.lib.asu.edu/lib/asulib-ebooks/reader.action?docID=4834260&ppg=2&c=RVBVQg
12. "Thinking Critically About and Researching Algorithms" by Rob Kitchin

Image Sources

1. https://illumin.usc.edu/algorithmic-symphonies-how-spotify-strikes-the-right-chord/
2. https://www.technologyreview.com/2024/08/16/1096276/spotify-algorithms-music-discovery-ux/
3. https://www.digitalmusicnews.com/2024/06/12/my-spotify-ad-campaign/
4. https://www.techgropse.com/success-story/spotify
5. https://cdn.dribbble.com/userupload/36956477/file/original-42f0604cd4ca4c4b03802bbdcbfed96b.png?resize=752x&vertical=center
6. https://alchetron.com/Payola
7. https://www.proserveit.com/blog/introduction-to-data-analysis

Thank You!